Graph-Feature-Enhanced Selective Assignment Network for Hyperspectral and Multispectral Data Classification

Wei Li, *Senior Member, IEEE*, Junjie Wang, Yunhao Gao, Mengmeng Zhang, Ran Tao, *Senior Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*— Due to rich spectral and spatial information, the combination of hyperspectral and multispectral images (MSIs) has been widely used for Earth observation, such as wetland classification. However, mining of meaningful features and effective fusion of multisource remote sensing data are still urgent problems to be solved. In this article, graph-feature-enhanced selective assignment network (GSANet) is proposed. On the one hand, a graph feature extraction module (GFEM) is designed to extract topological structure information and combine with the rich spectral–spatial information. In particular, the features obtained by convolution are first mapped to the graph feature space, and the graph convolution operation is used to achieve propagation between nodes for preserving topological structure information. Moreover, to reduce the difference of graph features resulting from the mapping function and better explore the complementary properties of multisource data, a novel graph fusion strategy-graph dependence fusion is designed. A transition graph is generated to enhance the association and interaction between different graph features, so as to avoid the information loss caused by simple fusion operation. On the other hand, a selective feature assignment module (SFAM) is developed to adaptively assign weights to different discriminative features. SFAM assigns weights to different features to selectively emphasize informative features and suppress less useful ones. Extensive experiments are conducted on two multisource remote sensing datasets, and the improvement of at least 1.27% and 0.98% compared to other state-of-the-art work demonstrates the superiority of the proposed GSANet.**

***Index Terms*— Graph feature extraction module (GFEM), hyperspectral and multispectral classification, multisource remote sensing data, selective feature assignment module (SFAM).**

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I. Introduction

With the development of Earth observation system, more and more wetland remote sensing data are available. Among them, hyperspectral images (HSIs) provide a wide spectral range and plenty of spectral information [1]–[3]. Different from HSIs, multispectral images (MSIs) have higher spatial resolution with more spatial contexture information [4]. However, because the annotation of remote sensing data is time-consuming with heavy workloads, how to obtain more robust features from the data has become the key. In addition, simple concatenation or summation operations makes it difficult to exploit the complementary advantages of hyperspectral and multispectral data, and hence, an effective fusion strategy is essential. In recent years, numerous classification methods have been proposed for wetland classification [5]–[9]. In [10], a novel hierarchical object-based random forest (RF) algorithm was proposed for distinguishing different wetland classes. In [11], the support vector machine (SVM) has been proven to be an effective approach. Liu *et al.* [12] combined multiple end-member spectral mixture analysis and RF to classify Yellow River Delta Wetland. In [13], a novel feature weighting method by integrating the Fisher linear discriminant analysis and physical interpretation was proposed for improving the classification accuracy of land cover. In [14], a feature sequence-based 3-D convolutional neural network (CNN) was designed, and then, a correction algorithm was introduced for wetland classification from MSIs.

Although these classification models mentioned above have made significant progress, the classification of single-source data is still unsatisfactory due to the spectral similarity of wetland vegetation. Therefore, multisource remote sensing data, due to its rich complementary information, is gradually drawn the attention. The ensuing problem is how to effectively fuse multisource data, and the general strategy can be divided into data fusion, feature fusion, and decision fusion. Data fusion is a fundamental fusion strategy that is usually performed directly on the acquired raw data, emphasizing the fusion of different image information on the basis of image elements [15]. In [16], object-based machine learning classifiers and ensemble analysis techniques were used to combine with spectral information from HSI and LiDAR-derived measures. In [17], a multimodal data fusion

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algorithm has been proposed to amplify the impact of an existing training set by linking labeled data to unlabeled data via unsupervised techniques. In [18], the Gram–Schmidt algorithm was utilized to fuse high-resolution SAR and optical images, and then, SVM was used for supervised classification. However, data redundancy and excessive computation time are still inevitable in data fusion.

Unlike data fusion, feature fusion aims to combine features extracted from multisource data, which retains enough semantic information while effectively reducing data redundancy. In [19], since the ultimate goal is to classify wetlands, several other features are used in addition to the spectral information to assess the separability of wetland classes, including various spectral indices. In [20], a two-tunnel CNN is designed to extract spectral–spatial features from HSI and then combine them with features extracted from LiDAR using the cascade block. In [21], a hierarchical random walk network (HRWN) was developed to capture boundary localization and spatial consistency. Zhang *et al.* [22] proposed an unsupervised feature extraction framework for collaborative classification of multisource remote sensing data. In [23], the combination of extinction profiles and CNN features enables the model to benefit from multilevel features, thus improving the classification accuracy. In [24], a generalized graph-based fusion method was developed to couple dimension reduction and feature fusion of the spectral information and morphological profiles.

As for decision fusion, it is the process of feeding features into multiple classifiers and optimizing the probabilistic output to obtain the final classification result. In [25], the spectral features, spatial features, elevation features, and graph fused features were fed into SVM classifiers, after which the four classification maps were fused through weighted majority voting. In [26], a novel decision fusion strategy was proposed to combine both linear and nonlinear features. In [27], an ensemble multiple kernel active learning framework was proposed to incorporate different types of features, after which decision fusion was implemented to make the final decision. In [28], feature fusion and decision fusion were simultaneously used to integrate heterogeneous features, which were extracted by two coupled CNNs.

In general, most of the methods described above require the use of HSI in combination with other data. This is because HSI is rich in spectral information that has the potential to distinguish between different classes. However, when vegetations with similar spectral information are encountered, HSI alone is not sufficient to achieve satisfactory performance [29]–[31]. At this point, MSI with higher spatial resolution provides more spatial contextual details to complement this deficiency. In this article, HSI from GaoFen-5 (GF-5) and MSI from Sentinel-2 (Sen-2) are utilized for joint classification. Although there have been significant advances in the problem of multisource data classification, it still requires large quantities of abstract features to train robust supervised classifiers, which are often difficult to acquire. Besides, the imbalance of feature expression still hinders the process of feature fusion. In order to solve the above two challenges, this article considers the following two aspects: 1) extract more meaningful features from multisource data and 2) assign different weights to different features to better utilize the complementary information.

Thus, a novel hyperspectral and multispectral data classification method named graph-feature-enhanced selective assignment network (GSANet) is proposed. First, spectral feature and spatial feature are extracted by two CNN-based feature extractors. Then, a graph feature extraction module (GFEM) is developed in combination with CNN to obtain more semantic features. In particular, the graph relational learning block and graph dependence fusion are designed to obtain the topological structure information. Finally, to make full use of the complementary information, an effective selective feature assignment module (SFAM) is designed to assign weights to different features, thus emphasizing meaningful features while ignoring the interference of useless features. Superior performance on two real multisource remote sensing datasets demonstrates the effectiveness of the proposed method.

In summary, the main contributions of this article are summarized as threefold.

1) Considering the complicated environmental composition and the mixture of vegetation, a novel hyperspectral and multispectral data classification framework named GSANet is constructed. The topological structure information hidden in multisource data and the complementarity are fully explored.
2) A GFEM is well-designed to capture the topology structure information implicit in multisource data. Since sample collection is extremely difficult in real scenarios, how to extract more effective features from samples is important in classification tasks. To solve this issue, this article uses graph convolutional networks to obtain the topological structure information among the ground objects so as to enhance the representation features.
3) Since extracted features from multisource data are different, simple concatenation or summation operations destroy the balance, an SFAM is developed to adaptively fuse hyperspectral and multispectral information. In particular, hyperspectral features, graph features, and multispectral features are assigned different weights so that their useful features are selectively emphasized and useless features are suppressed.

The remainder of this article is organized as follows: Section II presents the details of the proposed GSANet. Section III provides the experimental results together with the corresponding analysis and discussion. Finally, conclusions are drawn in Section IV.

II. Proposed GSANet Framework

In order to extract more useful features and better exploit the complementary benefits, we aim at utilizing graph convolutional networks to obtain the topological structure information and selectively assign weights to classification features from hyperspectral and multispectral data. Fig. 1 shows an overview of the proposed GSANet. The proposed model consists of three parts: spectral–spatial feature extraction network (SSEN), GFEM, and SFAM. First, spectral and spatial features are extracted through two branches of CNN with different

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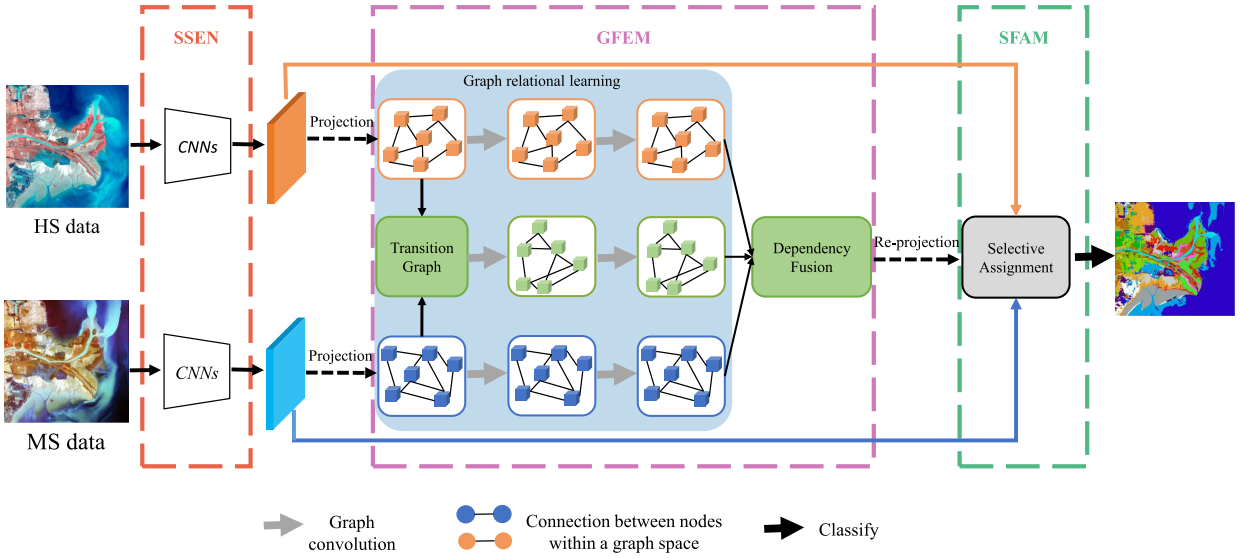


Fig. 1. Schematic illustration of the proposed GSANet. First, the patches from HSI and MSI are fed into SSEN to obtain the corresponding features. Then, they are projected into a graph space to learn the topological structure information through GFEM. After that, the graph feature and previous features are sent to SFAM to obtain enhanced classification features. Finally, the features are given to the classifier to generate the classification result.

parameters. Next, the extracted features are projected into a graph space and the proposed GFEM is used to obtain the topological structure information. Finally, an effective SFAM is developed to enhance the interaction of multisource data.

A. Spectral–Spatial Feature Extraction Network

Due to the efficiency and effectiveness of typical CNN, this article uses two CNNs to extract the spectral–spatial information from HSI and MSI separately. Given a HSI $I_H \in R^{c_1 \times h \times w}$ and an MSI $I_M \in R^{c_2 \times h \times w}$, c_1/c_2 is the number of spectral channels and h and w are the height and width of the image, respectively. Since the contextual information is critical for robust feature representation, spatial patches centered at position (i, j) with patch size r are fed into the SSEN to obtain the corresponding hyperspectral feature X_H and multispectral features X_M, where $0 < i < h$ and $0 < j < w$. This network consists of two branches of 2-D-CNNs with three convolutional layers, which are trained with the training samples, followed by a batch normalization (BN) layer and an activation layer.

For HSI and MSI, the spectral–spatial features are extracted as

$$\begin{aligned} \text{ConvBN}(c_1/c_2, 256) &\Rightarrow \text{ConvBN}(256, 128) \\ &\Rightarrow \text{ConvBN}(128, 128) \end{aligned} \tag{1}$$

where $\text{ConvBN}(c, c')$ is a convolutional layer followed by a batch norm (BN) and an activation layer and c and c' represent the number of channels of input and output data, respectively.

B. Graph Feature Extraction Module

Recently, many deep learning-based methods have been proposed for remote sensing data classification, which aims to use the spectral–spatial features to discriminate different classes. Nevertheless, these features are insufficient to accurately classify complex scene and indentify their classes [32], [33]. Therefore, how to reveal more meaningful features has become the key to improve the classification performance. We aim to use a GFEM to capture topological structure information that is not accessible to CNNs. It is divided into two parts. The first part uses graph convolution to perform relational learning for learning topological structure information. The second part is a graph dependence fusion block, which is used to enable the interaction and thus better exploit the complementary merits of multisource data.

1) Graph Relational Learning: Given feature tensors from the SSEN, graph relational learning is employed. To construct the graph, we project the features into high-level graph space. First, a graph as $G = (V, E)$ is defined, where V denotes the vertices and E denotes the edges. Then, the extracted features $X_{H,M} \in R^{C \times H \times W}$ are projected to high-level graph space $G \in R^{N \times D}$, where C is the dimension of the feature vector, H and W are the height and width of the feature vector, respectively, N represents the number of nodes, which corresponds to the number of categories in the scene, and D is the dimension of nodes.

Given the graph space representations, we leverage graph convolution [34] to propagate information over representation G of all nodes with matrix multiplication to obtain the evolved feature space G^p

$$G^p = \sigma_r(A \cdot G^{p-1} \cdot W + b) \quad p = 1, 2, 3 \tag{2}$$

where $W \in \mathbb{R}^{D \times D}$ and b are the weight matrix and bias, respectively, σ_r is the $ReLU$ function, and p is the number of layers in graph convolution network. The node adjacency weight matrix A is learnable and utilized to capture the relation between different classes. To better capture topology structure information, the adjacency matrix A is randomly initialized and continuously updated during learning. To sufficiently

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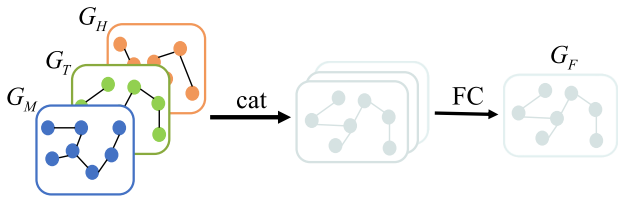


Fig. 2. Illustration of dependence fusion between three graph feature representations.

propagate information between nodes, the graph convolution is performed three times.

2) Graph Dependence Fusion: Due to the different feature focus of heterogeneous data, the graph feature spaces obtained through the mapping function reflect some differences. In order to better exploit the complementary merits of multisource data and reduce the difference of graph features, a fusion strategy that can enhance the interaction of heterogeneous data is necessary. The most straightforward solution is concatenation or element-wise summation. Nevertheless, these practices ignore feature associations and interactions from heterogeneous data. To solve these problems, a novel fusion block is designed.

In this fusion module, in order to better explore the complementary advantages between hyperspectral and multispectral data and avoid the difference of graph features caused by the mapping function, a transition graph G_T is first constructed, which is utilized to enhance the feature associations and interaction. Let $G_H \in \mathbb{R}^{N\times D}$ denote the hyperspectral graph feature and $G_M \in \mathbb{R}^{N\times D}$ denote the multispectral graph feature. G_T is formulated as

$$G_T = \alpha(G_H + F \cdot G_M \cdot W_M) + (1-\alpha)(G_M + F^T \cdot G_H \cdot W_H) \tag{3}$$

where α is a balance parameter, which is learned during the training process and $W_M, W_H \in \mathbb{R}^{D\times D}$ are weight matrices. $F = f_{i,j}, i, j = [1, N]$ is a fusion matrix, which is employed to better combine heterogeneous graph spaces. The node adjacency weight $f_{i,j}$ is calculated as

$$f_{i,j} = \cos(v_i^H, v_j^M) \tag{4}$$

where $\cos(v_i^H, v_j^M)$ is a function that measures the similarity between v_i^H and v_j^M, v_i^H is the ith node of hyperspectral graph feature space, and v_j^M is the jth node of multispectral graph feature space. When the similarity between v_i^H and v_j^M is high, the corresponding $f_{i,j}$ is larger, and thus, more information is combined from the other graph feature. Conversely, if v_i^H and v_j^M are very different, the corresponding adjacency weight between them is small, thus keeping more of their own characteristics and avoiding information confusion caused by the difference.

Subsequently, dependence fusion is designed to fuse G_H, G_T, and G_M, as shown in Fig. 2. After generating the transition graph via (4), the three types of graph features are concatenated in the feature dimension, after which the feature dimension is recovered to D using the fully connected (FC) layer to generate the fused graph feature G_F. It is formulated as

$$G_F = FC(\text{con}(G_H, G_T, G_M)) \tag{5}$$

where con$(\cdot)$ represents the concatenation operation and $FC()$ is the FC layer. This block learns associations between heterogeneous data by constructing a transition graph space as opposed to direct concatenation, thus better exploiting the complementary merits of multisource data.

Finally, the fused graph feature space representation is utilized to boost the classification characterization of spectral–spatial features. In particular, another mapping function is used to reproject G_F to features X_G. The whole implementation details of GFEM are provided in Algorithm 1.

Algorithm 1 Workflow of the Proposed GFEM

Input: Spectral-spatial features X_H and X_M.
Output: A couple of graph features X_G.

1: Apply mapping function to project spectral-spatial features to high-level graph space, obtaining the corresponding graph feature representation G_H and G_M.
2: Construct the transition graph G_T: $G_T = \alpha(G_H + F \cdot G_M \cdot W_M) + (1-\alpha)(G_M + F^T \cdot G_H \cdot W_H)$
3: **for** p = 1, 2, 3 **do**
4: Get the evolved feature G_H^p, G_M^p and G_T^p:

$$G_H^p = \sigma_r(A \cdot G_H^{p-1} \cdot W + b)$$
$$G_M^p = \sigma_r(A \cdot G_M^{p-1} \cdot W + b)$$
$$G_T^p = \sigma_r(A \cdot G_T^{p-1} \cdot W + b)$$

5: **end for**
6: Fuse the evolved feature through graph dependence fusion block to get the fused graph features representation G_F.
7: Apply another mapping function to re-project G_F to original feature space, obtaining X_G.

3) Plug-and-Play Feature Enhancement Module: As shown in Fig. 1, the proposed GFEM is embedded into CNN-based models. On the one hand, it does not change the size of feature maps. The reprojected image features remain the same size as input, and hence, it can be embedded after any convolution layer. On the other hand, the characterization of the final classification features is enhanced by the addition of the topological structure information extracted by GFEM.

C. Selective Feature Assignment Module

Considering that the features extracted from multisource data have different emphases, the most conventional methods for feature fusion, including concatenation or summation, only provide limited expressive power or even disrupt the balance of feature representation. Therefore, how to adaptively assign weights to heterogeneous features so as to make full use of complementary information is a key.

Recently, attention mechanism has been incorporated into image interpretation, which brings great improvement in performance [35]–[37]. One of the most influential methods is SENet [38], which considers the channel attention

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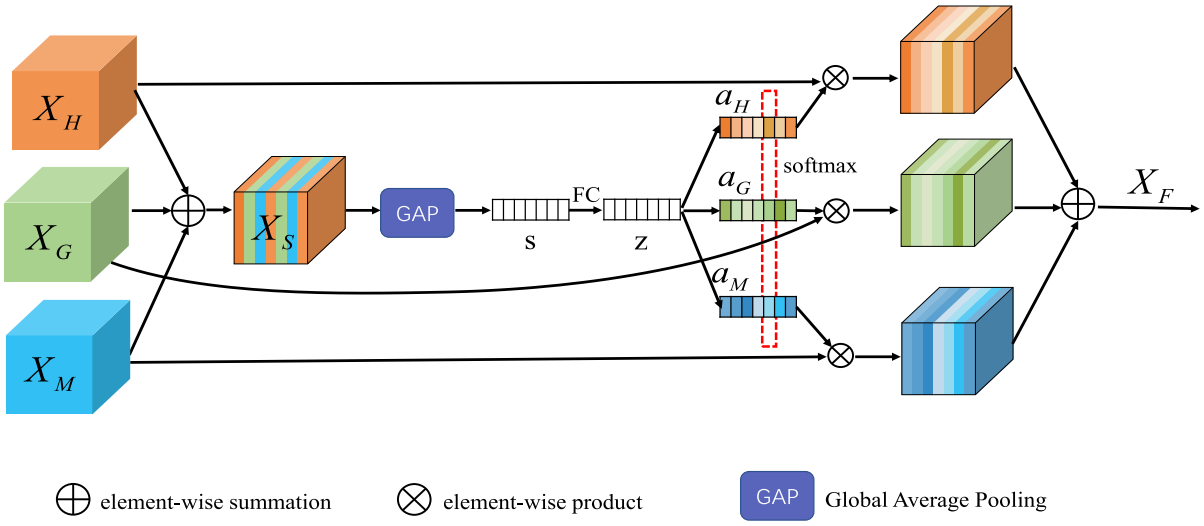


Fig. 3. Schematic of the proposed SFAM.

into the focus. Following that, several works incorporating other application scenarios have been proposed. Among them, SKNet [39] adaptively adjusts its receptive field size based on multiple scales of input information. However, when revisiting the attention mechanism of SKNet, it still follows the squeeze-to-excitation operation to obtain cross-channel interactions. Although this operation is widely used in attention modules, there are some studies [40], indicating that dimension reduction negatively affects correlation across channels. To overcome this deficiency, a selective feature assignment network is developed to fuse heterogeneous features.

The illustration of SFAM is shown in Fig. 3. Let $X_H \in \mathbb{R}^{C_H \times H \times W}$ denote the features from HSI, $X_M \in \mathbb{R}^{C_M \times H \times W}$ denote the features from MSI, and $X_G \in \mathbb{R}^{C_G \times H \times W}$ denote the features reprojected from the graph. Here, H and W indicate the height and width of the feature maps, respectively, and C_H, C_M, and C_G mean the number of channels. First, heterogeneous features are integrated from three branches via an element-wise summation

$$X_S = X_H + X_G + X_M. \tag{6}$$

Then, global average pooling (GAP) is utilized to obtain channel-wise statistics $s \in R^C$, which is calculated as

$$s = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_S(i, j). \tag{7}$$

Subsequently, an FC layer is applied to create a compact feature vector $z \in R^C$ that guides the feature selection. Unlike the squeeze operation in SKNet, z does not perform dimension reduction, which was proven to facilitate the learning of effective channel attention [40].

Next, the feature vector z passes through FC layers and the softmax function is applied to adaptively assign weights to heterogeneous features. Finally, the classification feature is obtained by fusing different branches with attention weights

$$X_F = a_H \cdot X_H + a_G \cdot X_G + a_M \cdot X_M \tag{8}$$

where the three attention vectors a_H, a_G, and a_M are generated by passing feature vector z through three FC layers.

III. Experimental Results and Discussion

A. Dataset Description and Evaluation Criteria

In order to verify the validity of the proposed method, two multisource datasets consisting of GF-5 and Sen-2 are utilized for experimental analysis. In particular, the spatial resolution of GF-5 is 30 m, and Sen-2 has a 10-m ground sample distance (GSD). In order to sample HSI from GF-5 to 10-m GSD, bilinear interpolation is used to match the resolution of MSI from Sen-2. Besides, the ground truth maps are also acquired with 10-m GSD. The spectral resolution of GF-5 is 5 nm in VNIR (400–1000 nm) with 150 bands and 10 nm in SWIR (1000–2500 nm) with 180 bands. It should be noted that necessary data preprocessing (such as geometric corrections, coregistration, and atmospheric correction) have been done [41].

The first dataset is Yellow River Estuary dataset. As shown in Fig. 4, the Yellow River Estuary Wetland dataset was captured in the Yellow River Delta in northeastern Shandong Province, China, and has the size of 3555 × 4026 pixels. The HSI and MSI were collected on November 1, 2008 and November 3, 2018, respectively. The Yellow River estuary Wetland is rich in natural resources and ecological diversity, with large areas of phragmites, Suaeda salsa, and complex mixed areas. Since the ground truth map is extremely important for accurate classification, as shown in Fig. 5, the ground class labels of the region of interest (ROI) are manually annotated carefully by integrating field investigation and expert knowledge. The training and testing sets are manually annotated by experts with rich knowledge in field trips and photograph interpretation, in which the geographical proximity and spacing are fully considered without overlapping regions [42], the details of which are listed in Table I.

The second one is Yancheng (YC) dataset. The YC wetland is located in the east of Jiangsu Province in China, as shown in

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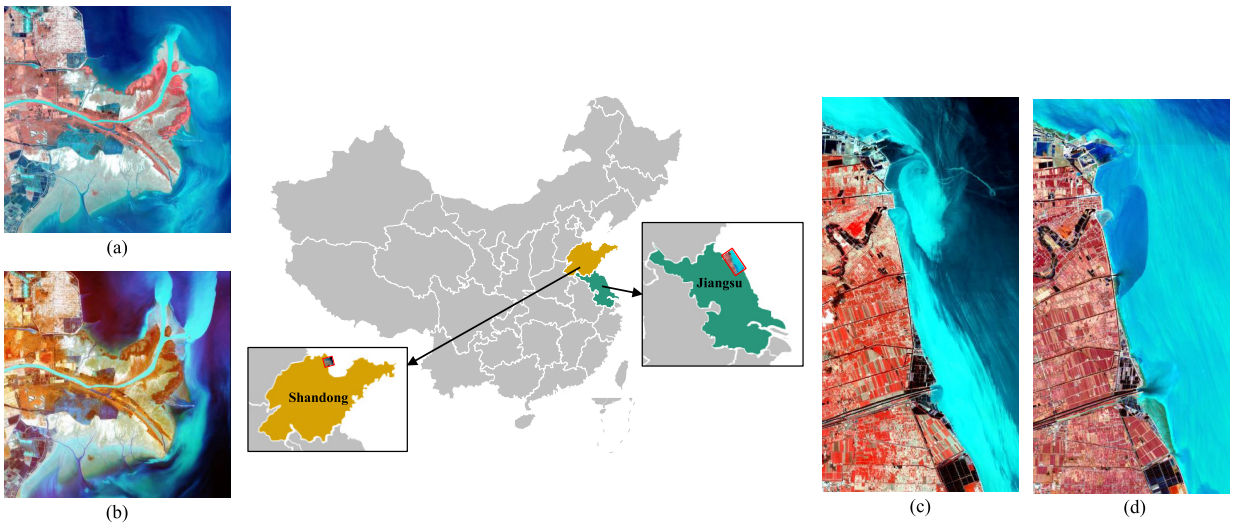


Fig. 4. Coastal wetland datasets. (a) HSI captured by GF-5 in Yellow River Estuary. (b) MSI captured by Sen-5 in Yellow River Estuary. (c) HSI captured by GF-5 in YC. (d) MSI captured by Sen-5 in YC.

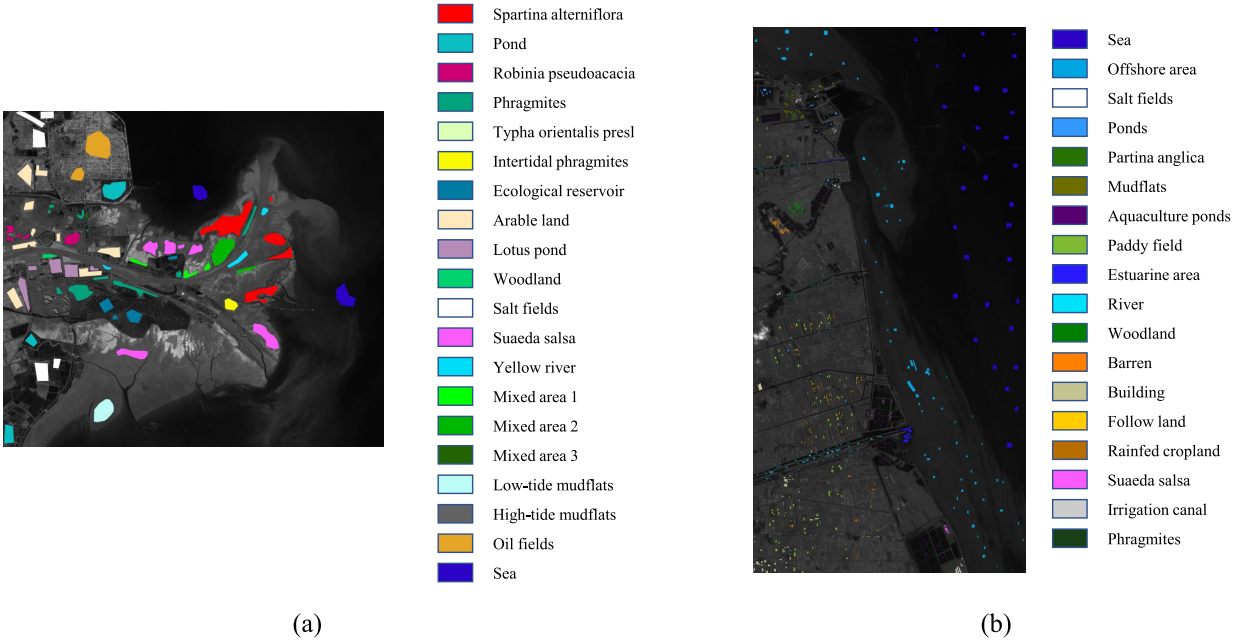


Fig. 5. ROI and ground truth maps for two datasets. (a) Yellow River Estuary dataset. (b) YC dataset.

Fig. 4. The size of YC dataset is 3525×1755 pixels, which was captured on April 4, 2019 and March 29, 2019. YC wetland contains a diverse range of vegetation types, numerous internal rivers, and rich biological resources, providing a variety of valuable resources for the surrounding residents. As shown in Fig. 5, the annotation of the true labels of the ROI is completed through a large amount of prior knowledge and combined with high spatial resolution images. Table I lists the number of training and testing samples that are manually annotated and divided from the annotated ROI based on the results of the field trips.

Besides, three quantitative evaluation indicators, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC), are used to evaluate the experimental results. The KC is computed as

$$\mathrm{KC} = \frac{P_o - P_e}{1 - P_e} \times 100\% \tag{9}$$

where P_o is the sum of the number of correctly classified samples of each class divided by the total number of samples, which equals the OA

$$P_e = \frac{t_1 \times p_1 + t_2 \times p_2 + \cdots + t_c \times p_c}{n \times n} \tag{10}$$

where $t_i, i = 1, 2, \ldots, c$ represents the number of true samples of class i and p_i is the number of samples predicted as class i.

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES IN TWO DATASETS

Yellow River Estuary Data Set				Yancheng Data Set			
Class		Number of Sample		Class		Number of Sample	
No.	Name	Training	Testing	No.	Name	Training	Testing
1	Sea	77	49841	1	Sea	316	21239
2	Ponds	102	62615	2	Ponds	36	1701
3	Phragmites	113	69232	3	Phragmites	24	1164
4	Salt fields	126	78173	4	Salt fields	24	966
5	Woodland	22	4903	5	Woodland	54	1881
6	Suaeda salsa	155	97294	6	Suaeda salsa	22	743
7	Spartina alterniflora	230	140411	7	Offshore area	190	14102
8	Robinia pseudoacacia	45	26221	8	Partina anglica	15	732
9	Typha orientalis presl	9	315	9	Mudflats	38	2374
10	Intertidal phragmites	27	12703	10	Aquaculture ponds	46	2321
11	Ecological reservoir	63	35226	11	Paddy fields	113	7375
12	Arable land	162	98462	12	Estuarine area	42	2433
13	Lotus pond	99	58554	13	River	59	2632
14	Yellow river	27	16840	14	Barren	36	1350
15	Mixed area 1	27	14649	15	Building	57	4677
16	Mixed area 2	79	49722	16	Fallow land	44	2062
17	Mixed area 3	21	1226	17	Rainfed cropland	48	1788
18	Low-tide mudflats	63	36535	18	Irrigation canal	18	324
19	High-tide mudflats	36	16988				
20	Oil fields	126	73091				
Total		1609	943102	Total		1182	69864

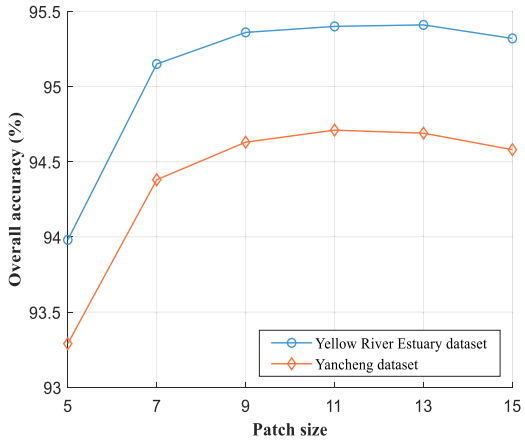


Fig. 6. Relationship between OA and patch size.

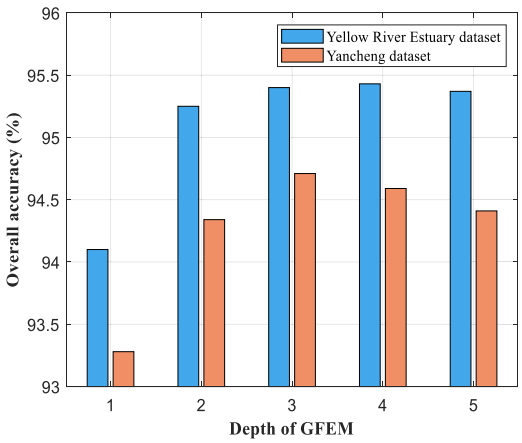


Fig. 7. Relationship between OA and the depth of GFEM.

B. Parameter Analysis of the Proposed Method

1) Analysis of the Patch Size: The size of image patch r mentioned in Section II-A is an important parameter, which controls the spatial contextual information contained in the input data. To verify the effect of different sizes r on the final classification results, we set $r = [5, 7, 9, 11, 13, 15]$, with larger r corresponding to more spatial contextual information contained in the corresponding input data. As shown in Fig. 6, the overall accuracy rises as the block size increases from 5 to 11, after which the performance does not improve significantly or even decreases. The reason is that when the patch size is small, the spatial contextual information around the input data is not fully explored. With the increase in size, the information becomes richer, which leads to the improvement of the classification results. However, once a certain size is reached, further expansion will only introduce additional distracting information, which not only affect the classification results but also increase the computational consumption. Therefore, in the subsequent experimental setup, the patch size of the input data was set to 11×11 pixels.

2) Analysis of the Depth of GFEM: As mentioned in Section II-B, multiple graph convolution operations are used to capture the topological structure information in the data, and the depth of GFEM corresponds to the number of graph convolution. Experiments are carried out to find the

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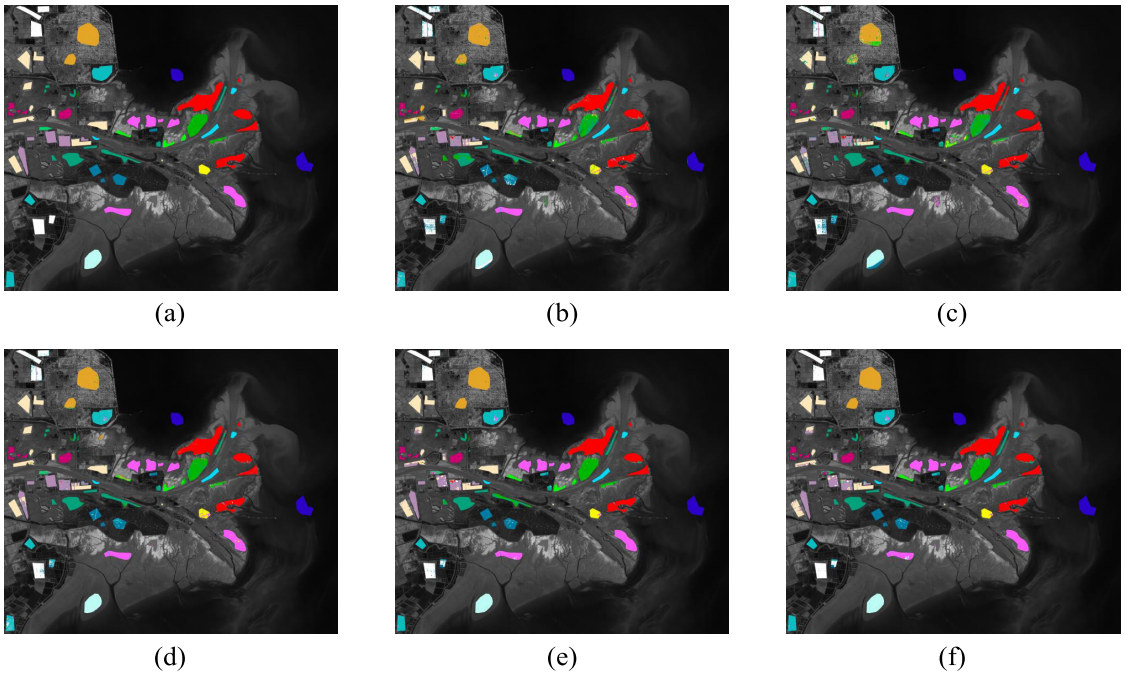


Fig. 8. Visualization results of different classification methods on the Yellow River Estuary dataset. (a) Ground truth map. Result of (b) SVM (86.50%), (c) LBP-ELM (86.60%), (d) two-branch CNN (92.67%), (e) DFINet (94.12%), and (f) GSANet (95.40%).

appropriate depth of the GFEM. The depth p is set to 1–5 to evaluate the classification performance, and the OA is used as the final evaluation metric. From Fig. 7, the trend of two datasets is slightly different. When $p < 3$, the value of OA increases sharply with the increase of depth. However, when $p > 3$, the increase of depth leads to the oversmoothing of the output features, and thus, the curve of Yellow River Estuary dataset first rises slightly and then remains stable, and the OA of the YC dataset decreases. As an intrinsic low-pass filter, the graph convolution gradually homogenizes the features of different nodes as the increase of convolution time. Therefore, due to the comprehensive consideration of classification performance and module complexity, the depth of GFEM is finally set to 3.

C. Classification Performance and Discussion

To validate the performance of the proposed method on multisource data classification, the method is compared with the following relevant methods.

1) *SVM [43]:* A radial basis function kernel is used to implement SVM, which is utilized for multisource data classification.
2) *LBP-ELM [44]:* The whole process is divided into two steps. In the first step, local binary pattern (LBP) is used to extract effective features, and then, extreme learning machine (ELM) is utilized to realize data classification.
3) *Two-Branch CNN [20]:* In two-branch CNN, two branches are employed for HSI and MSI feature extraction, after which the obtained features are combined for multisource data classification.
4) *DFINet [9]:* In DFINet, a deep-wise cross attention model is proposed to extract the self-correlation and cross correlation from feature pairs. Note that in [9], DFINet has been designed for the classification task with 30-m spatial resolution, whereas in this article, for a fair comparison, experiments are carried out with 10-m spatial resolution.

In the following experiments, we use GF-5 to represent the results obtained with HSI only and Sen-2 to represent results obtained with MSI only. Also, GF-5 + Sen-2 denotes the classification results obtained through the combination of HSI and MSI. Both visual and quantitative analyses have been done in order to prove the classification results of the different methods more intuitively and accurately.

1) Results on Yellow River Estuary Dataset: In this section, all experiments are conducted on the Yellow River Estuary dataset. In order to compare the classification performance more fairly, the training and test samples of all methods are listed in Table II. Fig. 8 shows the visualization of the final classification results of different classification methods. The classification results of the SVM and LBP-ELM contain a large number of noisy points within some regions, which is a reflection of misclassification. The two deep learning-based comparison methods have less noise points in the region, which is because deep learning can better learn abstract features, but there is still misclassification at the boundary of the different classes. In contrast, the results generated by GSANet contain the least noise points and still produce better classification results at the boundaries of the different classes. This is because the representation ability of classification features is improved by introducing GFEM. To reveal the discrepancies in the results of the different classification methods more clearly, two typical regions have been specifically selected for further analysis, as shown in Fig. 9. The differences

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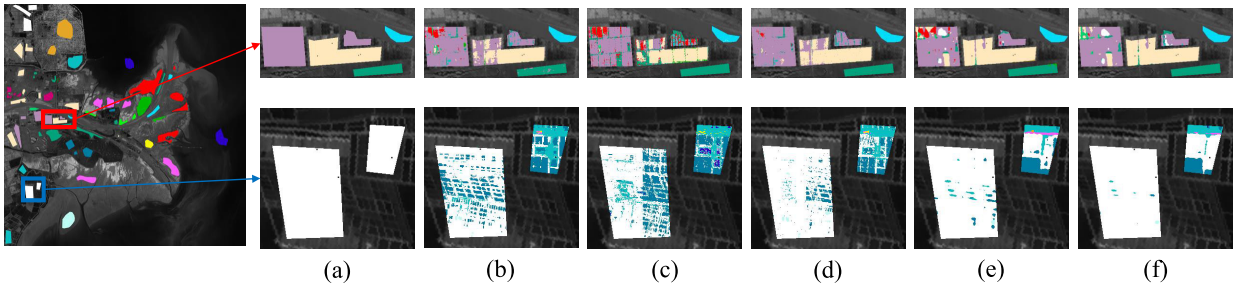


Fig. 9. Visualization results of typical areas in the Yellow River Estuary dataset with different classification methods. (a) Ground truth map. Result of (b) SVM, (c) LBP-ELM, (d) two-branch CNN, (e) DFINet, and (f) Result of GSANet.

TABLE II
CLASSIFICATION ACCURACY (100%) OBTAINED BY DIFFERENT APPROACHES ON THE YELLOW RIVER ESTUARY DATASET

Class	SVM [43]			LBP-ELM [44]			Two-branch CNN [20]			DFINet [9]	GSANet
	GF-5	Sen-2	GF-5+Sen-2	GF-5	Sen-2	GF-5+Sen-2	GF-5	Sen-2	GF-5+Sen-2	GF-5+Sen-2	GF-5+Sen-2
1	74.91	99.27	99.45	99.65	99.93	**100.00**	**100.00±0.00**	98.84±0.64	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	80.23	86.84	90.12	81.46	84.96	95.36	94.45±1.35	92.19±5.15	94.05±3.39	94.22±1.54	**96.93±0.35**
3	78.10	83.84	72.59	85.14	83.07	83.45	78.66±12.09	82.66±11.00	83.94±9.38	89.19±6.27	**96.33±1.33**
4	87.52	76.33	75.35	88.41	77.64	76.22	89.90±2.01	85.84±2.31	88.45±1.55	**90.81±1.22**	90.59±0.96
5	74.14	76.62	71.21	67.25	23.68	63.31	86.78±3.67	**95.25±4.43**	84.80±1.68	63.51±1.24	80.77±1.97
6	81.13	98.14	97.34	84.57	98.51	98.22	97.93±1.47	95.21±2.08	**99.23±0.51**	99.14±0.21	98.22±0.27
7	89.26	95.09	94.88	82.61	99.06	**98.98**	96.71±0.59	97.75±0.45	97.09±0.61	97.60±0.28	97.39±0.35
8	85.55	95.74	95.57	80.51	98.96	97.73	98.60±1.39	96.50±2.14	98.89±1.08	**99.59±0.32**	99.13±0.11
9	98.73	99.68	99.68	35.87	**100.00**	99.37	**100.00±0.00**	97.88±1.09	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	74.07	73.46	77.57	29.11	78.96	80.43	92.32±2.52	89.23±8.25	86.38±4.68	90.89±1.54	**93.88±0.41**
11	77.85	76.70	78.93	**94.42**	94.03	91.32	87.73±2.34	87.59±3.25	88.64±2.36	89.13±1.53	90.46±1.34
12	89.69	90.24	85.83	78.51	84.54	87.20	91.96±5.83	83.23±9.12	88.98±3.43	89.48±1.61	**92.38±1.77**
13	79.66	84.10	85.96	69.73	68.41	73.53	75.65±1.27	84.69±2.27	85.03±1.25	**86.89±0.80**	86.34±0.96
14	99.43	**100.00**	**100.00**	96.17	65.84	97.20	96.01±0.64	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
15	61.38	58.64	61.86	47.31	40.43	44.80	79.01±4.98	84.75±6.15	79.16±2.11	80.68±1.48	**86.87±1.67**
16	82.25	55.10	64.30	67.64	39.58	68.85	84.55±4.40	65.29±5.35	85.66±1.82	89.46±0.97	**92.91±1.21**
17	69.06	14.94	69.47	71.41	66.48	86.59	**90.76±3.07**	89.45±5.18	79.93±18.11	90.11±7.72	84.19±10.30
18	98.89	82.75	79.37	97.09	83.96	80.38	**100.00±0.00**	**100.00±0.00**	97.49±1.56	**100.00±0.00**	**100.00±0.00**
19	89.92	88.83	92.82	39.87	70.28	71.90	**99.21±0.29**	90.33±2.68	95.07±0.44	98.97±0.16	99.19±0.12
20	84.16	86.33	92.65	75.71	84.00	75.32	97.30±0.80	95.40±2.22	97.60±0.64	**99.64±0.16**	99.61±0.16
OA	84.08	86.49	86.50	80.79	83.94	86.60	92.05±0.38	90.10±0.65	92.68±0.26	94.06±0.17	**95.33±0.40**
AA	82.80	81.13	84.25	73.62	77.12	83.51	91.88±0.70	90.60±0.96	91.52±0.29	92.47±0.84	**94.26±0.95**
KC	82.65	85.31	85.32	79.14	82.52	85.43	91.36±0.41	89.25±0.69	92.04±0.28	93.54±0.18	**94.92±0.44**

between the different classification results can be seen more clearly in Fig. 9, and the result map generated by GSANet has the least noise points and the most accurate classification performance, which is closest to ground truth map. This also fully reflects the effectiveness of the proposed GSANet for the problem of classifying coastal wetland from multisource data.

From Table II, the different classification methods achieved relatively good classification results on the Yellow River Estuary dataset. Compared with other methods, the proposed GSANet achieves the best performance on all three evaluation indicators. Through the comparison with SVM and LBP-ELM, OA achieves an improvement of at least 8.73%, which is very impressive. Compared with the two deep learning-based methods (two-branch CNN and DFINet), it also achieved 2.65% and 1.27% improvements, respectively. From the perspective of data types, when HSI or MSI is used alone, the classification result is obviously not as good as the combination of them, which also proves the complementary merits of multisource data. In particular, when using SVM as a classifier, the data input of GF-5 + Sen-2 achieved the best results on all three classification metrics. Although the OA values are close to the combined data when MSI is used as input, the classification performance of individual classes is extremely poor, resulting in an obvious decrease in AA values. The results of LBP-ELM and two-branch CNN are similar. Although single-source data has achieved competitive results with multisource data, multisource data fusion is still the best choice. GSANet distributes the weights of multisource data through the introduction of SFAM, makes better use of the complementarity of multisource data, and achieves the best classification results in most cases.

2) Results on the Yancheng Dataset: For the purpose of visual analysis, the final classification maps are shown in Fig. 10. Due to the huge scale of the scene and in order to better reveal the classification result, two representative locations are chosen to demonstrate the differences between the various classification methods in greater detail, as shown in Fig. 11. In the first region, the classification results of the SVM divided almost all pixels belonging to the aquaculture

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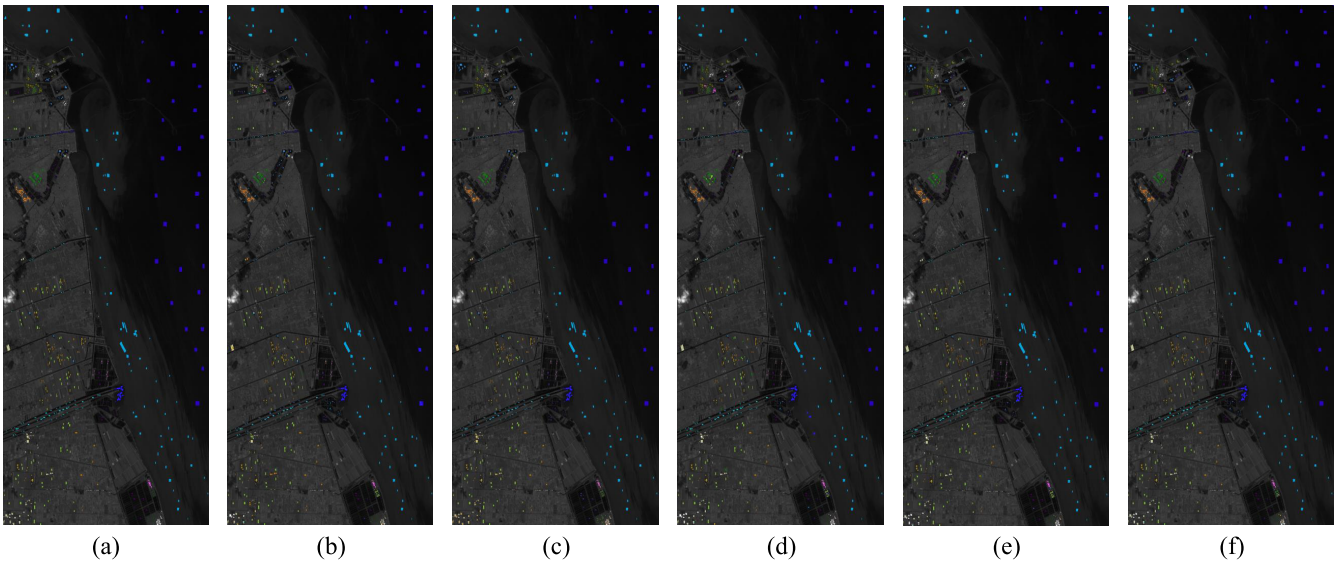


Fig. 10. Visualization results of different classification methods on the YC dataset. (a) Ground truth map. Result of (b) SVM (86.21%), (c) LBP-ELM (89.34%), (d) two-branch CNN (91.79%), (e) DFINet (93.30%), and (f) GSANet (94.71%).

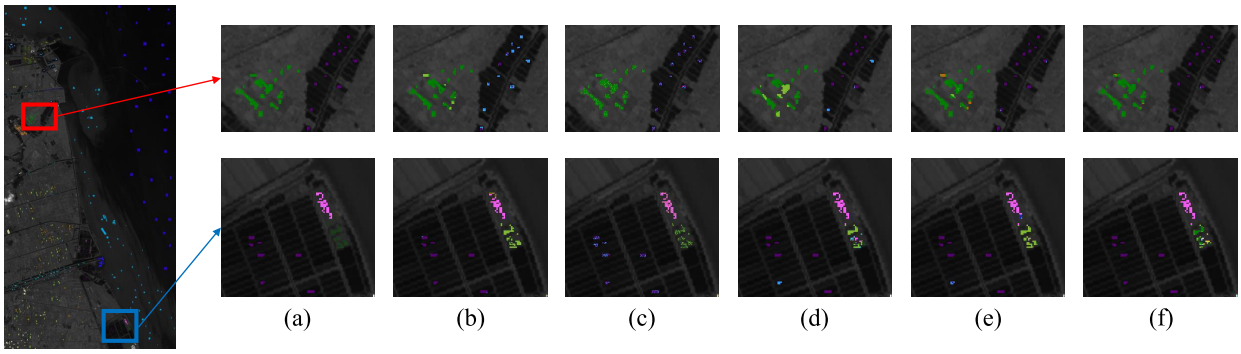


Fig. 11. Visualization results of typical areas in the YC dataset with different classification methods. (a) Ground truth map. Result of (b) SVM, (c) LBP-ELM, (d) two-branch CNN, (e) DFINet, and (f) GSANet.

ponds into ponds, which is due to the low discrimination of shallow features between the two classes, leading to a substantial number of misclassifications. In the case of LBP-ELM, the presence of noisy points in the classification results has corrupted the classification performance. For two-branch CNN and DFINet, they wrongly classify part of the woodland as either paddy field or rainfed cropland. In contrast, GSANet produces the results closest to the real distribution. Similarly, in the second region, SVM still classifies almost all pixels belonging to phragmites as paddy field due to the similarity of the water conditions. There are still a large number of noisy points throughout the whole region in the result of LBP-ELM, which severely restricts the improvement of classification accuracy. In contrast, in phragmites distribution areas, two-branch CNN and DFINet misclassify lots of pixels into paddy field due to the similarity of their spectral patterns. When it comes to the proposed GSANet, the introduction of topological structure information minimizes the probability of misclassification to some extent, and the overall classification result is most similar to the ground truth map.

Table III shows the classification results of the YC dataset produced by the different classification methods with different inputs. First, GSANet has achieved the best results in all three most important evaluation indicators. In terms of data combination, similar to the Yellow River Estuary dataset, the best classification results achieved by SVM, LBP-ELM, and two-branch CNN are with multisource data inputs. In particular, the accuracies of the three methods have increased at least 0.53%, 2.18%, and 1.83%, after utilizing multisource data. According to the results of two datasets, multisource data fusion has tremendous significance for the task of remote sensing data classification. From the perspective of classification methods, traditional machine learning methods, including SVM and LBP-ELM perform inconsistently in terms of classification. It is difficult for them to capture complicated feature representations through shallow models and do not classify well on some classes, resulting in lower AA values. Deep learning-based methods (two-branch CNN and DFINet) can capture high-level features through multiple hidden layers to better distinguish various classes.

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TABLE III
CLASSIFICATION ACCURACY (100%) OBTAINED BY DIFFERENT APPROACHES ON THE YC DATASET

Class	SVM [43]			LBP-ELM [44]			Two-branch CNN [20]			DFINet [9]	GSANet
	GF-5	Sen-2	GF-5+Sen-2	GF-5	Sen-2	GF-5+Sen-2	GF-5	Sen-2	GF-5+Sen-2	GF-5+Sen-2	GF-5+Sen-2
1	92.39	81.64	93.43	96.23	95.62	99.18	99.95±0.04	96.63±2.44	99.92±0.07	99.91±0.06	**99.97±0.02**
2	**69.49**	58.85	66.90	63.14	59.61	62.90	64.54±1.11	59.28±0.87	65.94±1.19	66.71±1.15	65.78±0.69
3	60.82	48.97	63.75	61.94	51.55	64.26	58.45±5.61	53.29±5.60	63.75±1.70	63.46±3.19	**65.49±2.41**
4	77.74	70.29	76.40	70.39	75.05	70.60	65.46±3.23	55.62±3.35	**90.72±1.52**	86.16±1.60	71.91±1.19
5	88.20	65.23	86.23	90.22	84.74	90.70	52.53±10.13	59.81±11.91	84.95±6.92	91.85±1.56	**99.27±0.72**
6	80.35	41.72	79.54	81.83	53.57	76.45	67.38±3.18	48.95±5.89	83.09±2.29	**95.90±1.51**	89.05±0.23
7	99.88	42.14	99.87	**100.00**	61.02	**100.00**	**100.00±0.00**	77.42±2.50	98.74±0.77	**100.00±0.00**	**100.00±0.00**
8	**100.00**	**100.00**	**100.00**	78.01	92.90	97.54	84.11±2.24	78.96±2.88	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
9	63.86	37.57	62.43	80.92	42.00	**92.00**	55.67±1.91	57.05±1.73	65.94±1.09	71.47±1.09	80.60±1.22
10	66.18	53.55	55.67	80.05	65.23	77.51	84.75±2.49	74.52±2.91	85.52±2.28	**88.61±1.37**	86.53±1.86
11	95.32	69.67	97.97	96.65	75.97	98.79	96.52±1.46	84.54±1.85	94.75±1.36	96.20±0.58	**98.15±0.18**
12	95.36	77.35	95.44	**97.21**	73.90	98.40	90.78±0.57	80.90±0.57	93.09±0.43	95.04±0.31	92.70±0.45
13	52.51	74.39	54.60	62.54	77.20	63.18	96.88±0.34	91.93±1.04	96.01±0.32	**99.34±0.21**	99.25±0.64
14	97.41	56.74	97.70	88.41	75.93	99.41	80.94±1.15	79.46±1.06	82.02±1.22	99.73±0.14	**99.95±0.03**
15	31.84	60.96	36.37	30.79	66.90	31.54	66.65±2.28	80.09±1.05	62.77±1.01	82.78±1.40	**84.88±1.01**
16	98.45	34.63	**98.55**	73.76	33.56	86.66	78.10±1.59	49.24±5.90	73.88±1.85	61.56±1.31	76.62±1.46
17	95.08	59.68	**95.13**	89.82	61.02	89.93	61.65±1.22	65.16±1.87	78.97±1.31	80.07±0.55	83.17±0.51
18	62.04	40.12	71.60	53.09	55.56	69.75	43.93±4.12	51.65±4.66	65.84±2.64	**99.90±0.05**	98.35±1.16
OA	85.68	63.61	86.21	87.16	74.41	89.34	88.85±1.33	80.94±2.43	90.68±1.28	93.35±0.30	**94.33±0.28**
AA	79.27	59.64	79.53	78.11	66.74	81.60	74.89±1.71	69.14±1.44	82.55±1.41	87.71±0.50	**88.43±0.58**
KC	83.16	56.60	83.77	84.85	69.51	87.39	86.75±1.59	77.30±1.64	88.96±1.21	92.11±0.36	**93.27±0.33**

Compared to traditional methods, the accuracy improvement of at least 1.34% also proves its effectiveness. The proposed GSANet enhances the discrimination of the final classification features by introducing GFEM to obtain the implicit topological structure information in the data and employs SFAM to adaptively fuse different features, allowing the full exploration of complementary property between multisource data. This is corroborated by the accuracy improvement of 0.98%–8.12%.

3) Analysis of Number of Training Samples: It is widely acknowledged that training samples play an important role in deep learning. However, since labeling remote sensing data is a time-consuming and labor-intensive process, requiring the assistance of professional knowledge, for this reason, in the task of hyperspectral and multispectral data classification, the effect of different numbers of training sample on the experimental results needs to be considered. Therefore, in this section, the relationship between OA and the number of training samples on different datasets has been investigated. Here, [20%, 40%, 60%, 90%] of the samples from the training set are randomly selected as the new training set. The classification performance obtained on different datasets by using different numbers of training samples is shown in Figs. 12 and 13. As can be seen from the curve that as the number of training samples increases, the OA value first rises sharply and then becomes stable. This is because a small amount of training samples is not only insufficient for the training of the network but also leads to overfitting. However, due to the introduction of topology structure information, the proposed GSANet can better make up for the impact of insufficient training samples, so it also achieves satisfying results with little training samples.

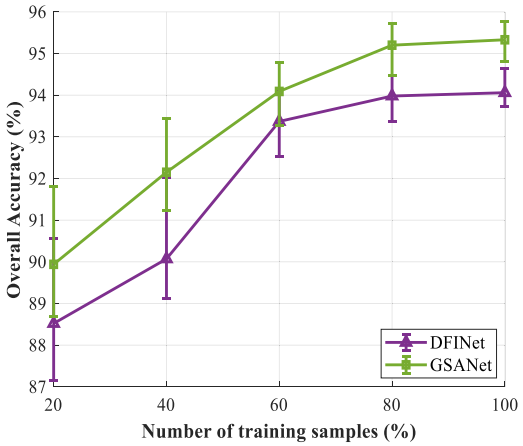


Fig. 12. Relationship between OA and the number of training samples on the Yellow River Estuary dataset.

D. Ablation Studies and Running Time Comparisons

Next, a series of ablation experiments is conducted to verify the effectiveness of the proposed modules in improving the classification accuracy, and the specific experimental results are listed in Table IV. In order to reflect the role of different modules more comprehensively, the experiments are roughly divided into three parts as follows.

1) Verify the effect of the two proposed modules (GFEM and SFAM) on the final results. By comparing #1 with #5, the designed GFEM enhances the representation ability of the final classification features by capturing the topological structure information hidden in the data, which greatly improves the performance of the model.

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TABLE IV
ABLATION STUDIES FOR THE PROPOSED GSANET ON TWO DATASETS

Class	GFEM			SFAM		OA (%) on different dataset	
	Graph Relational Learning	Transition Graph	Dependency fusion	Unsqueeze	Squeeze	Yellow River Estuary dataset	Yancheng dataset
1	✓	✓	✓	✓		95.40	94.71
2		✓	✓	✓		93.00	92.87
3	✓		✓	✓		94.86	94.25
4	✓	✓		✓		95.31	94.61
5				✓		91.52	91.29
6	✓	✓	✓		✓	95.30	94.58
7	✓	✓	✓			95.08	94.34

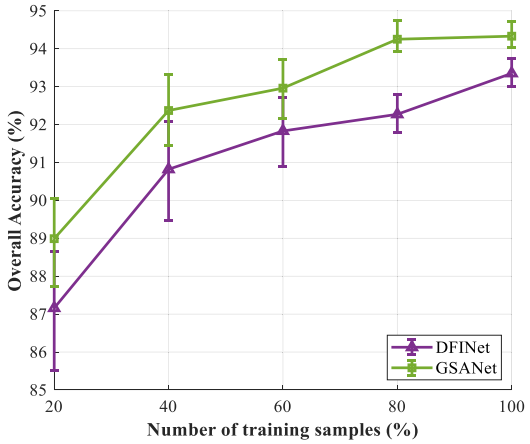


Fig. 13. Relationship between OA and the number of training samples on the YC dataset.

Second, through the comparison between #1 and #7, it is proven that the proposed SFAM plays an unsubstitutable role in feature assignment, where the fusion of different data can be better handled by assigning weights to different classification features.

2) The function of each submodule in the proposed GFEM is verified by comparing the results of three experiments. #2 represents the learning process without graph relational learning, that is, without graph convolution operations. The comparison with #1 is that multiple feature learning is necessary. #3 means learning directly in the graph spaces without generating transition graph. From the results, the final accuracy has decreased. This is because multisource data have a diverse feature focus, making it hard to integrate the graph space effectively without feature transition. #4 denotes that, rather than using the fusion process shown in Fig. 2, the features are summed directly, which inevitably leads to the loss of some information and reduces the classification accuracy.
3) To evaluate the negative effect of the dimension reduction mentioned in Section II-A, a squeeze operation is used to replace the original FC layer. The experimental results from #6 illustrate that compared with the originally designed module, dimension reduction brings information loss. To sum up, on the one hand, the proposed GSANet acquires topological structure information through the proposed GFEM, which not only improves the discrimination of classification features but also integrates multisource information in graph space; on the other hand, SFAM better handles meaningful information by assigning weights to different classification features and reduces information loss by discarding squeezing operations. Through the combination of two modules, GSANet has achieved a highly competitive performance on the task of multisource data classification of coastal wetlands.

Fig. 14 visualizes the feature distribution under different variants to more intuitively highlight the influence of the proposed two modules on feature classification. As shown in Fig. 14(a), the original features extracted by SSEN are severely confused, resulting in the lack of discrimination between different classes, making it difficult to distinguish them. As shown in Fig. 14(b), when GFEM is added to the network, more information is extracted and utilized, so there is a significant improvement in the learning capability of the whole network. The introduction of GFEM has improved the separation degree between classes and enhances the discrimination of them, providing power for further improvements in network performance. Finally, the feature distribution under the complete network structure is shown in Fig. 14(c). Compared to the variants, the feature distribution is more reasonable, the degree of aggregation within the class is higher, and the interval between classes is more obvious, demonstrating the usefulness of the proposed modules.

Time consumption is one of the major factors that restricts the application of deep learning-based methods in remote sensing data classification tasks. Table V reports the running times of the proposed GSANet and other methods. All experiments are carried out on a desktop PC with an Intel Core i7-9700 processor and an Nvidia GeForce RTX 3070 GPU.

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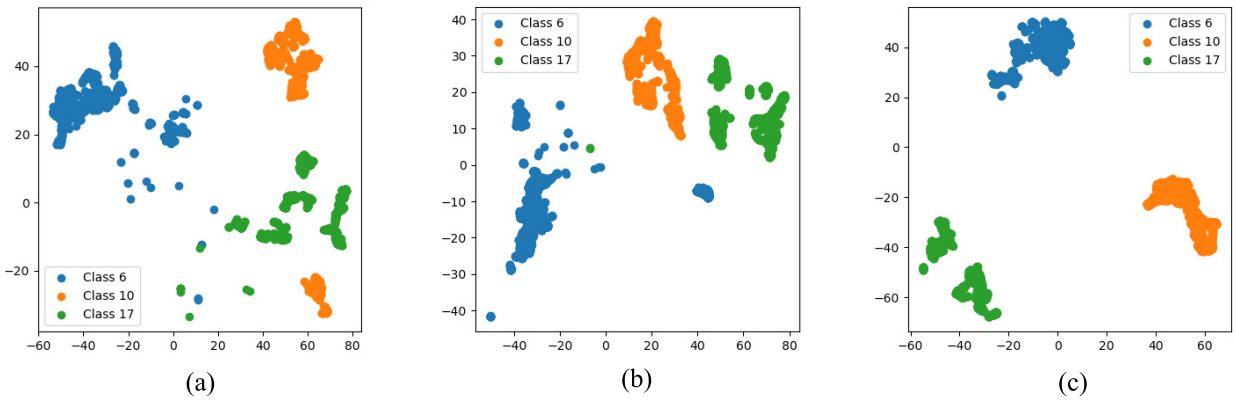


Fig. 14. Visualization of the feature distribution from some selected classes on the YC dataset. Features generated by (a) SSEN, (b) GFEM, and (c) GSANet.

TABLE V
RUNNING TIME COMPARISONS OF DIFFERENT CLASSIFICATION METHODS

Method	Yellow River Estuary dataset		Yancheng dataset	
	Running time (s)	OA (%)	Running time (s)	OA (%)
SVM	38.97	86.50	13.71	86.21
LBP-ELM	5.61	86.60	2.05	89.34
Two-branch CNN	449.85	92.67	173.47	91.79
DFINet	196.94	94.12	169.81	93.30
GSANet	387.70	95.40	160.98	94.71

As can be seen in Table V, the traditional machine learning methods (SVM and ELM) have lower running times. Compared to the methods based on deep learning, their model structures are simpler and require fewer parameters to be optimized, making them more computationally efficient. Two-branch CNN has a deeper network structure, requiring data processing and feature extraction. What is more, there are lots of parameters that need to be optimized, and thus, it consumes more time. In contrast, the depth of DFINet network is relatively smaller, and its optimized network structure is more efficient than two-branch CNN. Because GSANet introduces a graph convolution network, it needs time to perform graph relational learning, but the network structure is shallower and requires fewer parameters to optimize, so the running time is within an acceptable range. Considering the running time and classification accuracy, GSANet is a competitive candidate in multisource data classification methods.

IV. CONCLUSION

In this article, an effective GSANet was proposed for hyperspectral and multispectral data classification by fully exploiting the meaningful features. In particular, a new feature extractor, GFEM, was developed to obtain the topology structure information. Moreover, a novel graph fusion strategy was designed to explore the complementary advantages between multisource data and reduce the variance of graph features resulted from the mapping function. The discrimination of samples has been improved by fusing graph features with spectral–spatial features. In addition, the SFAM was utilized for feature fusion, which takes better advantage of the complementary merits of hyperspectral and multispectral data than the simple feature summation or concatenation. The 1.27% and 0.98% gains achieved on the Yellow River Estuary dataset and the YC dataset also reveal the effectiveness of the proposed GSANet.

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